

**Fourth Semester B.Tech. (Electronics and Communication Engineering)
Examination**

ANALOG AND DIGITAL COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1.
 - (a) An audio frequency signal $10 \sin 2\pi \times 500t$ used to amplitude modulates a carrier of $50 \sin 2\pi \times 10^5t$. Determine modulation index, bandwidth, transmission efficiency, total power delivered to load of 600Ω , Sketch the frequency spectrum. 5(CO1)
 - (b) A 20 MHz carrier is modulated by 400 Hz modulating signal. The carrier voltage is 5V and maximum deviation is 10KHz. Write down standardized mathematical equation for FM and PM wave. 5(CO1)
 2.
 - (a) Illustrate the effect of White Noise in Double Sideband Full Carrier (DSBFC) system. 5(CO2)
 - (b) Two resistors $20 \text{ K}\Omega$ and $50 \text{ K}\Omega$ are at room temperature 290 K. By considering bandwidth of 100 KHz, determine thermal noise for the conditions for each resistor, for two resistances in series, for two resistances in parallel. 5(CO2)
 3.
 - (a) Derive an expression for SNR in PCM for analog signal input. 5(CO3)
 - (b) Illustrate the importance of non uniform quantization. Explain in detail Adaptive delta modulation. 5(CO3)

4. Describe in detail QPSK Transmitter and Receiver with mathematical expression. Draw waveforms for QPSK Transmitter if input is 10110001. 10(CO4)
5. A Convolution Code is described by $X1 = [101]$; $X2 = [111]$; Draw the convolution encoder and state diagram. If the received sequence for above encoder is 000110, decode the sequence using Viterbi decoding algorithm. 10(CO4)
6. Derive an expression for SNR and Probability of error for
- (1) Amplitude Shift keying
 - (2) Phase shift keying. 10(CO5)

